ADDITIONAL FILE 1: SUPPLEMENTAL METHODOLOGY

A.1. WOMEN'S HEALTH INITIATIVE:

Study design and population: The Women's Health Initiative is a large prospective study of postmenopausal women aged 50–79 years at enrollment (1993–1998), including both clinical trial and observational components. For the present analysis, we restricted the sample to women free of diabetes at baseline, with valid anthropometric measurements, complete covariate data, and more than 12 months of follow-up. Women with body mass index (BMI) <18.5 kg/m² or missing key variables were excluded.

Exposure, outcome, and covariates: BMI was calculated from measured height and weight and categorized using standard cut points (normal, overweight, obesity classes I–III). Waist circumference (WC) was measured at baseline using standardized protocols and categorized using BMI-specific thresholds. Joint BMI–WC categories were created to represent combined adiposity phenotypes, with normal BMI and low WC as the reference group. Incident type 2 diabetes was defined based on annual self-reported use of insulin or oral hypoglycemic medications during follow-up. Covariates included demographic, socioeconomic, and lifestyle characteristics measured at baseline, selected based on prior literature and harmonized with external validation cohorts.

Statistical analysis: The analytic sample was randomly split into training (70%) and test (30%) datasets to develop and internally validate prediction models. Time-to-event analyses were conducted using Cox proportional hazards regression to estimate hazard ratios and 95% confidence intervals for type 2 diabetes risk across BMI and joint BMI–waist circumference categories. Unadjusted and fully adjusted models were examined. Incidence rates were estimated using Poisson methods, and Kaplan–Meier curves with log-rank tests were used to assess differences in cumulative incidence. Model performance was evaluated using Harrell's c-statistics, and effect modification by age, race, and ethnicity was assessed using interaction terms.

Additional considerations: Proportional hazards assumptions were evaluated using residual-based diagnostics and graphical methods. Multiple testing was addressed using Bonferroni correction where appropriate. Analyses were conducted using SAS version 9.4 and Stata version 19, and complete-case analysis was used after evaluating patterns of missingness. This framework enabled consistent estimation and validation of type 2 diabetes risk models based on overall and central adiposity in postmenopausal women.

A.2. HEALTH AND RETIREMENT STUDY:

Study design and population: The Health and Retirement Study (HRS) is a nationally representative longitudinal study of U.S. adults aged ≥ 50 years. For this analysis, data from the RAND HRS longitudinal file (1992–2020) were merged with the 2022 Tracker file using a unique respondent identifier (HHIDPN). The analytic sample was restricted to women aged ≥ 50 years who were alive and interviewed in 2006, had valid anthropometric data, and were free of type 2 diabetes at baseline. Participants with BMI < 18.5 kg/m², missing key exposures or covariates, or insufficient follow-up were excluded. Follow-up extended through 2020, with administrative censoring on January 1, 2021.

Exposure, outcome, and measurements: BMI was calculated using measured or reported height and weight and categorized using standard clinical cut points (normal, overweight, and obesity classes I–III). WC was obtained from the 2006 Enhanced Face-to-Face (EFTF) assessment, where trained interviewers collected anthropometric measurements using standardized protocols. BMI-specific WC thresholds were applied to define central adiposity within each BMI category, and joint BMI–WC categories were constructed to capture combined adiposity phenotypes. Incident type 2 diabetes was defined using self-reported physician diagnosis or treatment (insulin or oral medications) across follow-up waves. Time-to-event variables were constructed using interview dates derived from tracker file month and year variables, allowing precise estimation of follow-up time.

Covariates: Covariates were selected based on prior literature and availability in HRS and were harmonized to align with WHI definitions. Demographic variables included age (continuous and categorized), race/ethnicity, and marital

status. Socioeconomic variables included education and household wealth, derived from RAND summary variables. Behavioral variables included smoking status (never, former, current), alcohol consumption (categorized by frequency), and physical activity based on self-reported frequency of moderate or vigorous activity. All covariates were defined at or prior to baseline to ensure temporal ordering relative to incident type 2 diabetes.

Statistical analysis: Time-to-event analyses were conducted using Cox proportional hazards models with attained age as the underlying time scale and delayed entry at baseline age. Follow-up time was calculated from baseline interview date to incident diabetes, death, loss to follow-up, or administrative censoring. Analyses accounted for the complex survey design using sampling weights, strata, and primary sampling units. Unadjusted and fully adjusted models were examined. Kaplan–Meier curves were generated to visualize cumulative incidence across BMI and BMI-specific WC categories, and differences were assessed using log-rank tests. Model discrimination was evaluated using Harrell's c-statistics. No stratified analyses by age or race/ethnicity were conducted to assess consistency of associations across subgroups, due to reduced sample size.

A.3. HEALTHY AGING IN NEIGHBORHOODS OF DIVERSITY ACROSS THE LIFE SPAN:

Study design and population: The Healthy Aging in Neighborhoods of Diversity across the Life Span (HANDLS) is a prospective cohort study of community-dwelling adults aged 30–64 years at baseline, designed to examine health disparities by race and socioeconomic status in an urban U.S. population. For the present analysis, we restricted the sample to women who had valid anthropometric data, complete covariate information, and were free of type 2 diabetes at baseline. Participants with BMI <18.5 kg/m² or missing key variables were excluded. Follow-up extended through the most recent available examination cycle, with time-to-event defined from baseline assessment to incident type 2 diabetes, death, loss to follow-up, or administrative censoring.

Exposure, outcome, and measurements: BMI was calculated from measured height and weight obtained during standardized clinical examinations and categorized using conventional cut points (normal, overweight, and obesity classes I–III). WC was measured by trained personnel using standardized protocols at the level of the iliac crest. BMI-specific WC thresholds were applied to define central adiposity within BMI strata, and joint BMI–WC categories were constructed to represent combined adiposity phenotypes. Incident type 2 diabetes was defined based on self-reported physician diagnosis, medication use, or clinical criteria where available, harmonized to match definitions used in WHI and HRS.

Covariates: Covariates were selected to align with WHI and HRS and included demographic, socioeconomic, and behavioral factors measured at baseline. Demographic variables included age and race. Socioeconomic status was captured using education and poverty status measures. Behavioral covariates included smoking status (never, former, current), and alcohol consumption, based on standardized questionnaires and diet quality as measured by HEI-2010 (2 24-hour recalls). Variables were harmonized across cohorts to ensure comparability of effect estimates. Physical activity was not available at wave 1 of HANDLS and therefore was not included among potential confounding covariates.

Statistical analysis: Associations between adiposity measures and incident type 2 diabetes were evaluated using Cox proportional hazards regression models. Follow-up time was defined from baseline to incident diabetes or censoring. Unadjusted and fully adjusted models were examined. Kaplan–Meier curves were generated to visualize cumulative incidence across BMI and BMI-specific WC categories, and differences were assessed using log-rank tests. Model performance was evaluated using Harrell's c-statistics. No stratified analyses by age or race were conducted to assess consistency of associations across subgroups, due to reduced sample size.